

# Alpilla 600 MW Combined Cycle Power Plant Plant Design Basis Report

|                   |                                                                                                                                                                                                                                                  |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Document number   | ALP-EPC-00000-GE-DBR-0001                                                                                                                                                                                                                        |
| Revision / status | Rev A - IFR, issued 2026-11-16                                                                                                                                                                                                                   |
| Originator        | Istanbul EPC Mühendislik ve Taahhüt A.Ş.                                                                                                                                                                                                         |
| Discipline / type | general / DBR Design basis report                                                                                                                                                                                                                |
| Review class      | approval (Basic Design Package)                                                                                                                                                                                                                  |
| Basis             | requirement:ER-01.01, requirement:ER-01.02, requirement:ER-01.03, requirement:ER-01.04, requirement:ER-01.05, requirement:ER-02.01, requirement:ER-02.02, requirement:ER-02.03, requirement:ER-02.05, requirement:ER-02.11, requirement:ER-03.01 |

| Rev | Purpose | Date       | Description                                           | Prepared | Checked          | Approved                |
|-----|---------|------------|-------------------------------------------------------|----------|------------------|-------------------------|
| A   | IFR     | 2026-11-16 | First issue for Owner approval (Basic Design Package) | EPCE     | EPC process lead | EPC engineering manager |

## 1. Purpose and scope

This report fixes the design basis of Alpilla 600 MW Combined Cycle Power Plant: the site, climatic, fuel, grid, water and environmental conditions, the performance and operating requirements and the guarantees against which the plant is designed. All values are taken from the project database; each value states its status (assumption, preliminary, confirmed) and its basis (Owner document, requirement, reference or decision). Discipline design criteria (process, mechanical, electrical, I&C, civil / structural, HVAC, HSE) are derived from this report.

## 2. Plant configuration

1 x 1 multi-shaft combined cycle: one gas turbine with generator, one three-pressure reheat heat recovery steam generator with SCR, one reheat condensing steam turbine with its own generator and a seawater-cooled surface condenser; both generators connected through generator step-up transformers to the 380 kV GIS switchyard. Main equipment (register):

| Tag     | Description                                      | System | Supply |
|---------|--------------------------------------------------|--------|--------|
| 00XJA10 | Emergency diesel generator 1,250 kVA             | 00XJA  | IEPC   |
| 10BAT10 | GT step-up transformer 560 MVA 400/21 kV         | 10BAT  | IEC    |
| 10HAA10 | Heat recovery steam generator, 3-pressure reheat | 10HAA  | IEC    |
| 10MBA10 | Gas turbine                                      | 10MBA  | IEC    |
| 10MKA10 | GT generator 560 MVA, H2-cooled                  | 10MKA  | IEC    |
| 20BAT10 | ST step-up transformer 270 MVA 400/15.75 kV      | 20BAT  | IEC    |
| 20MAA10 | Steam turbine, reheat condensing, 2 casings      | 20MAA  | IEC    |
| 20MKA10 | ST generator 270 MVA, TEWAC                      | 20MKA  | IEC    |

92 plant systems (KKS, ALP-EPC-00000-GE-PRC-0001), by category:

| Category             | Systems                                                                                                                                                                                                                                                                                                                        |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| auxiliary mechanical | 00QCA Chemical dosing, 00QEA Compressed air, 00QHA Auxiliary boiler, 00QJA Nitrogen supply, 00QJB Hydrogen and CO2 supply, 00QUA Sampling and SWAS, 10HSB Ammonia storage and supply                                                                                                                                           |
| civil structural     | 00GQA Storm water drainage                                                                                                                                                                                                                                                                                                     |
| common               | 00BXA Earthing and lightning protection                                                                                                                                                                                                                                                                                        |
| condensate feedwater | 10LAB Feedwater piping, 10LAC Boiler feed pumps, 10LAE HP attemperation spray water, 10LAF Reheat attemperation spray water, 20LCA Main condensate piping, 20LCB Condensate extraction pumps                                                                                                                                   |
| cooling              | 00PAA CW intake, screens and outfall, 00PAB CW piping, 00PAC CW pumps, 00PAH Condenser tube cleaning, 00PBA Electro-chlorination, 00PGA Closed cooling water                                                                                                                                                                   |
| electrical hv        | 00ACA 380 kV GIS switchyard, 10BAA GT generator busduct, 10BAC GT generator circuit breaker, 10BAT GT generator step-up transformer, 20BAA ST generator busduct, 20BAC ST generator circuit breaker, 20BAT ST generator step-up transformer                                                                                    |
| electrical mv lv     | 00BBA MV switchgear 10.5 kV, 00BFA LV switchgear 0.4 kV, 00BFT LV transformers, 00BLA Lighting and small power, 00BMA Emergency LV switchgear, 00BRA UPS, 00BTA Batteries and DC 220 V, 00XJA Emergency diesel generator, 10BBT GT unit auxiliary transformer, 20BBT ST unit auxiliary transformer                             |
| fire protection      | 00SGA Fire water system, 00SGC Deluge, spray and foam systems                                                                                                                                                                                                                                                                  |
| fuel gas             | 00EGA LDO unloading, 00EGB LDO storage, 00EGC LDO forwarding and treatment, 00EKA Gas receiving, filtration and fiscal metering station, 00EKC Fuel gas distribution to GT and auxiliary boiler                                                                                                                                |
| gas turbine          | 10MBA Gas turbine, 10MBL GT inlet air system, 10MBN GT liquid fuel and water injection, 10MBP GT fuel gas system incl. performance heater, 10MBU GT compressor washing, 10MBV GT lube oil system                                                                                                                               |
| generator            | 10MKA GT generator, 10MKC GT excitation and static frequency converter, 10MKG GT generator gas system (H2/CO2), 20MKA ST generator, 20MKC ST excitation                                                                                                                                                                        |
| hrsg                 | 10HAA HRSG pressure system, 10HNA HRSG flue gas ducting, 10HNE Stack 65 m with CEMS platform and silencer, 10HSA SCR catalyst and ammonia injection                                                                                                                                                                            |
| hvac                 | 00SAA HVAC                                                                                                                                                                                                                                                                                                                     |
| i and c              | 00CJA Plant DCS, 00CJG Safety instrumented systems (BOP), 00CKA Historian, performance monitoring and operator training simulator, 00CQA Continuous emission monitoring, 00CYA Telecom, CCTV, access control, security, 00SGY Fire detection and alarm, 10MBY GT control and protection (TCS), 20MAY ST control and protection |

| Category          | Systems                                                                                                                                                                                                                                                                         |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| other             | 00SMA Cranes and hoists                                                                                                                                                                                                                                                         |
| steam turbine     | 20MAA HP turbine, 20MAB IP/LP turbine, 20MAJ Condenser vacuum system, 20MAV ST lube oil system, 20MAW Gland steam system, 20MAX ST control fluid system                                                                                                                         |
| water steam cycle | 00LBG Auxiliary steam, 10HAN HRSG blowdown and drains (EPC tanks and headers), 10LBA Main steam piping, 10LBB Hot reheat piping, 10LBC Cold reheat piping, 10LBD LP steam piping, 20MAG Condensing system, 20MAN Turbine bypass stations and piping                             |
| water treatment   | 00GBA Seawater reverse osmosis, 00GCA Demineralisation, 00GHA Demineralised water storage and distribution, 00GKA Potable water, 00GMA Process drains and waste water collection, 00GNA Waste water treatment, 00GUA Sanitary waste water treatment, 00LDA Condensate polishing |

### 3. Site data

Site location, levels, loads and soil conditions for the civil and structural design; site class and seismic hazard per TBDY 2018.

| Parameter                                     | Value                                   | Condition                                                                                                          | Status    | Basis                                                                                          |
|-----------------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------|
| 100-year coastal still water level            | 0.9 m                                   | national datum; +1.40 m with 0.5 m sea-level rise; floor levels $\geq$ +1.90 m (ER-02.09); shoreline run-up +4.8 m | confirmed | SRC-OWNR-0002 Seawater quality and marine conditions, ER-02.09 Flood protection                |
| Atmospheric corrosivity category              | C5 (ISO 12944-2), durability H -        |                                                                                                                    | confirmed | ER-02.08 Corrosive coastal atmosphere                                                          |
| Basic wind speed                              | 30 m/s                                  | 10 min mean, 10 m, 50-year return                                                                                  | confirmed | ER-02.05 Wind, snow and rain                                                                   |
| Design rainfall intensity                     | 110 mm/h                                | 10 min, 25-year return                                                                                             | confirmed | ER-02.05 Wind, snow and rain                                                                   |
| Distance to heavy-lift port                   | 18 km by sea                            | 400 t quay crane                                                                                                   | confirmed | SRC-OWNR-0005 Site, existing infrastructure and tie-in points                                  |
| Ground snow load                              | 0.75 kN/m <sup>2</sup>                  |                                                                                                                    | confirmed | ER-02.05 Wind, snow and rain                                                                   |
| Groundwater level                             | 13.5 m a.s.l.                           | brackish, chloride 3,000-8,000 mg/l                                                                                | confirmed | SRC-OWNR-0008 Geotechnical baseline summary                                                    |
| Heavy transport road limit                    | 120 t gross                             | two bridges on the state road                                                                                      | confirmed | SRC-OWNR-0005 Site, existing infrastructure and tie-in points, ER-02.14 Access and transport   |
| Height datum                                  | TUDKA-99 orthometric heights (m a.s.l.) |                                                                                                                    | confirmed | SRC-EPCE-0011 Topographic survey point listing - plot, temporary area TP-A1 and coastal strip  |
| Plant grid origin in UTM zone 35N (E 0 / N 0) | E 541,250.000 m / N 4,532,480.000 m     | TUREF (ITRF96), south-west plot corner                                                                             | confirmed | SRC-EPCE-0011 Topographic survey point listing - plot, temporary area TP-A1 and coastal strip  |
| Platform level acceptance tolerance           | 0.15 m                                  | Owner-levelled platform +15.00 m                                                                                   | confirmed | DEC-OWNR-0002 Acceptance tolerance of the Owner-levelled platform                              |
| Plot dimensions                               | 400 x 300 m                             | 12.0 ha; usable about 10.6 ha                                                                                      | confirmed | SRC-OWNR-0005 Site, existing infrastructure and tie-in points, ER-02.01 Site location and area |
| Rotation plant grid to UTM grid               | -0.3281 deg                             | plant north = true north; grid convergence at site                                                                 | confirmed | SRC-EPCE-0011 Topographic survey point listing - plot, temporary area TP-A1 and coastal strip  |
| Scale factor plant grid to UTM                | 0.999612 -                              | combined UTM point scale and height factor                                                                         | confirmed | SRC-EPCE-0011 Topographic survey point listing - plot, temporary area TP-A1 and coastal strip  |
| Seismic design PGA (DD-2, 475-year)           | 0.45 g                                  | TBDY 2018, site-specific hazard study                                                                              | confirmed | ER-02.06 Seismic design                                                                        |
| Seismic site class                            | ZD (Vs30 = 240 m/s) -                   | TBDY 2018                                                                                                          | confirmed | SRC-OWNR-0008 Geotechnical baseline summary                                                    |
| Site grade level                              | 15 m a.s.l.                             |                                                                                                                    | confirmed | ER-02.01 Site location and area                                                                |

### 4. Ambient and climatic conditions

Design ambient conditions, Site Reference Conditions (SRC) for performance, salt-laden marine atmosphere and lightning data.

| Parameter                                         | Value   | Condition | Status    | Basis                         |
|---------------------------------------------------|---------|-----------|-----------|-------------------------------|
| Ambient temperature - maximum for full capability | 40 degC |           | confirmed | ER-02.02 Ambient design range |
| Ambient temperature - minimum for full capability | -8 degC |           | confirmed | ER-02.02 Ambient design range |

| Parameter                            | Value                | Condition                 | Status    | Basis                                                                              |
|--------------------------------------|----------------------|---------------------------|-----------|------------------------------------------------------------------------------------|
| Ambient temperature - survival range | -15 to +45 degC      |                           | confirmed | ER-02.02 Ambient design range                                                      |
| Annual mean temperature (measured)   | 14.9 degC            | 2015-2024 site mast       | confirmed | SRC-OWNR-0001 Meteorological and ambient air data                                  |
| Chloride deposition maximum          | 420 mg/m2/day        | ISO 9225, category S3     | confirmed | SRC-OWNR-0001 Meteorological and ambient air data, ER-02.10 Airborne salt and dust |
| Ground flash density                 | 2.4 flashes/km2/year | 26 thunderstorm days/year | confirmed | SRC-OWNR-0001 Meteorological and ambient air data, ER-02.12 Lightning              |
| Performance test ambient range       | 5 to 30 degC         |                           | confirmed | SRC-OWNR-0001 Meteorological and ambient air data, ER-02.11 Design ambient points  |
| PM10 maximum (event)                 | 260 ug/m3            | Saharan dust episodes     | confirmed | SRC-OWNR-0001 Meteorological and ambient air data, ER-02.10 Airborne salt and dust |
| Record maximum / minimum temperature | 39.6 / -11.4 degC    | 1960-2024                 | confirmed | SRC-OWNR-0001 Meteorological and ambient air data                                  |
| Sea-salt aerosol annual mean         | 14 ug/m3             | as NaCl                   | confirmed | SRC-OWNR-0001 Meteorological and ambient air data                                  |
| Sea-salt aerosol maximum (event)     | 165 ug/m3            | as NaCl, lodos storms     | confirmed | SRC-OWNR-0001 Meteorological and ambient air data, ER-02.10 Airborne salt and dust |
| SRC ambient dry-bulb temperature     | 15 degC              | Site Reference Conditions | confirmed | ER-02.03 Site Reference Conditions (SRC)                                           |
| SRC barometric pressure              | 1,011.5 mbar         | Site Reference Conditions | confirmed | ER-02.03 Site Reference Conditions (SRC)                                           |
| SRC relative humidity                | 70 %                 | Site Reference Conditions | confirmed | ER-02.03 Site Reference Conditions (SRC)                                           |
| Summer design dry bulb (0.4 %)       | 32.8 degC            | coincident RH 48 %        | confirmed | SRC-OWNR-0001 Meteorological and ambient air data, ER-02.11 Design ambient points  |
| Winter design dry bulb (99.6 %)      | -4.8 degC            | coincident RH 88 %        | confirmed | SRC-OWNR-0001 Meteorological and ambient air data, ER-02.11 Design ambient points  |

## 5. Fuels

Natural gas (main fuel) at the terminal point.

| Parameter                                      | Value               | Condition                                                                   | Status    | Basis                                                                                            |
|------------------------------------------------|---------------------|-----------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------|
| Booked gas capacity (firm)                     | 120,000 Sm3/h       | firm from the first gas date (TQ-040), 130,000 interruptible                | confirmed | SRC-OWNR-0003 Natural gas quality and supply conditions                                          |
| Exceptional minimum supply pressure            | 40 barg             | max 72 h/year, 24 h notice                                                  | confirmed | SRC-OWNR-0003 Natural gas quality and supply conditions                                          |
| Fuel gas superheat above dew points at GT skid | 28 K                | minimum                                                                     | confirmed | ER-03.06 Natural gas contaminant limits                                                          |
| Fuel LHV range                                 | 44.2 to 50.0 MJ/kg  |                                                                             | confirmed | ER-03.02 Natural gas specification                                                               |
| Gas supply pressure at terminal point          | 45 to 70 barg       |                                                                             | confirmed | ER-03.03 Gas supply conditions at terminal point                                                 |
| Gas supply temperature at terminal point       | 5 to 25 degC        |                                                                             | confirmed | ER-03.03 Gas supply conditions at terminal point                                                 |
| H2S + COS limit                                | 5 mg/Sm3            |                                                                             | confirmed | SRC-OWNR-0003 Natural gas quality and supply conditions, ER-03.06 Natural gas contaminant limits |
| Hydrocarbon dew point                          | -2 degC (1-70 barg) | maximum                                                                     | confirmed | SRC-OWNR-0003 Natural gas quality and supply conditions                                          |
| Hydrogen admixture capability                  | 5 vol %             | without modification; 10 vol % by future OEM retrofit (provisions included) | confirmed | ER-03.02 Natural gas specification                                                               |
| Mercury limit                                  | 1 ug/Sm3            | measured < 0.01                                                             | confirmed | SRC-OWNR-0003 Natural gas quality and supply conditions, ER-03.06 Natural gas contaminant limits |
| Methane content range                          | 85.0 to 98.5 mol %  |                                                                             | confirmed | SRC-OWNR-0003 Natural gas quality and supply conditions, ER-03.02 Natural gas specification      |

| Parameter                           | Value                            | Condition                                                     | Status    | Basis                                                                                            |
|-------------------------------------|----------------------------------|---------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------|
| Na + K, Pb, V in gas (network data) | < 0.005 (detection limit) mg/Sm3 | not guaranteed by network operator                            | confirmed | SRC-OWNR-0003 Natural gas quality and supply conditions, ER-03.06 Natural gas contaminant limits |
| Reference fuel LHV                  | 47.48 MJ/kg                      | reference composition, ISO 6976, combustion reference 25 degC | confirmed | ER-03.02 Natural gas specification                                                               |
| Total sulphur limit                 | 30 mg S/Sm3                      | measured typical 4, max 17                                    | confirmed | SRC-OWNR-0003 Natural gas quality and supply conditions, ER-03.06 Natural gas contaminant limits |
| Water dew point                     | -8 degC at 70 barg               | maximum                                                       | confirmed | SRC-OWNR-0003 Natural gas quality and supply conditions                                          |
| Wobbe index range                   | 47 to 52 MJ/Nm3                  |                                                               | confirmed | ER-03.02 Natural gas specification                                                               |

Light distillate oil (back-up fuel).

| Parameter                                 | Value                             | Condition                                     | Status    | Basis                                                                                                           |
|-------------------------------------------|-----------------------------------|-----------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------|
| LDO annual operating limit                | 500 h/year                        | back-up fuel status                           | confirmed | SRC-OWNR-0004 Secondary fuel: light diesel oil (LDO), SRC-OWNR-0007 Positive EIA decision - binding conditions  |
| LDO autonomy at base load                 | 72 h                              | two tanks of 50 %                             | confirmed | SRC-OWNR-0004 Secondary fuel: light diesel oil (LDO), ER-03.08 LDO unloading and storage                        |
| LDO density at 15 degC                    | 820 to 845 (840 for sizing) kg/m3 |                                               | confirmed | REF-CONS-0096 TS EN 590, SRC-OWNR-0004 Secondary fuel: light diesel oil (LDO)                                   |
| LDO flash point                           | 55 degC                           | minimum, EN ISO 2719                          | confirmed | SRC-OWNR-0004 Secondary fuel: light diesel oil (LDO)                                                            |
| LDO LHV                                   | 42.9 MJ/kg                        | range 42.6-43.2                               | confirmed | SRC-OWNR-0004 Secondary fuel: light diesel oil (LDO), ER-03.07 Light diesel oil specification and burner limits |
| LDO Na + K at GT inlet                    | 0.5 mg/kg                         | maximum; as delivered up to 1.0 (history 1.8) | confirmed | SRC-OWNR-0004 Secondary fuel: light diesel oil (LDO), ER-03.07 Light diesel oil specification and burner limits |
| LDO resupply capacity                     | 25 trucks/day                     | 30-36 m3 each                                 | confirmed | SRC-OWNR-0004 Secondary fuel: light diesel oil (LDO)                                                            |
| LDO sulphur                               | 10 mg/kg                          | maximum                                       | confirmed | SRC-OWNR-0004 Secondary fuel: light diesel oil (LDO), ER-03.07 Light diesel oil specification and burner limits |
| LDO vanadium / lead / calcium at GT inlet | 0.5 / 1.0 / 2.0 mg/kg             | maximum                                       | confirmed | SRC-OWNR-0004 Secondary fuel: light diesel oil (LDO), ER-03.07 Light diesel oil specification and burner limits |

## 6. Grid connection

Connection to the 380 kV transmission grid (TSO).

| Parameter                            | Value                          | Condition                                     | Status    | Basis                                  |
|--------------------------------------|--------------------------------|-----------------------------------------------|-----------|----------------------------------------|
| 380 kV voltage range                 | 360 to 410 kV                  | 342-420 kV for up to 30 min                   | confirmed | SRC-OWNR-0006 Grid connection data     |
| Connection capacity                  | 640 MW export                  | 40 MW import                                  | confirmed | SRC-OWNR-0006 Grid connection data     |
| Generator power factor range         | 0.85 lagging to 0.95 leading - | at rated active power                         | confirmed | ER-04.08 Reactive capability           |
| Grid connection voltage              | 380 kV                         | nominal                                       | confirmed | ER-11.01 Grid connection               |
| Grid frequency                       | 50 Hz                          | operating range 47.5 to 52.5 Hz per Grid Code | confirmed | ER-04.07 Frequency and voltage support |
| Short-circuit level at 380 kV (2029) | 12 (min) to 40 (max) kA        | future 50 kA (2035)                           | confirmed | SRC-OWNR-0006 Grid connection data     |
| Switchyard short-circuit rating      | 50 kA                          |                                               | confirmed | ER-11.01 Grid connection               |

## 7. Performance and operation

Performance requirements at SRC.

| Parameter                   | Value          | Condition          | Status    | Basis                    |
|-----------------------------|----------------|--------------------|-----------|--------------------------|
| Auxiliary power consumption | 2.2 % of gross | maximum, base load | confirmed | ER-04.10 Auxiliary power |
| Load ramp rate              | 35 MW/min      | minimum            | confirmed | ER-04.05 Load ramp rate  |

| Parameter                                   | Value           | Condition                  | Status    | Basis                               |
|---------------------------------------------|-----------------|----------------------------|-----------|-------------------------------------|
| Minimum environmental load                  | 40 % net output | maximum                    | confirmed | ER-04.06 Minimum environmental load |
| Net electrical output (guaranteed)          | 600 MW          | SRC, 380 kV terminal point | confirmed | ER-04.01 Net electrical output      |
| Net heat rate LHV (guaranteed)              | 5,860 kJ/kWh    | SRC                        | confirmed | ER-04.02 Net heat rate              |
| Net heat rate LHV at 60 % load (guaranteed) | 6,250 kJ/kWh    | SRC                        | confirmed | ER-04.03 Part-load efficiency       |

Operating regime and design life.

| Parameter                    | Value           | Condition                  | Status    | Basis                                     |
|------------------------------|-----------------|----------------------------|-----------|-------------------------------------------|
| Cold start time to base load | 180 min         | maximum                    | confirmed | ER-04.04 Start-up times                   |
| Design life                  | 30 years        |                            | confirmed | ER-01.03 Design life and operating regime |
| Hot start time to base load  | 45 min          | maximum                    | confirmed | ER-04.04 Start-up times                   |
| Operating hours per year     | 8,000 EOH/year  |                            | confirmed | ER-01.03 Design life and operating regime |
| Starts per year              | 250 starts/year | 50 cold, 100 warm, 100 hot | confirmed | ER-01.03 Design life and operating regime |
| Warm start time to base load | 90 min          | maximum                    | confirmed | ER-04.04 Start-up times                   |

## 8. Cooling water and water supply

Once-through seawater cooling; seawater quality at the intake; condenser design.

| Parameter                                               | Value            | Condition                             | Status    | Basis                                                                                 |
|---------------------------------------------------------|------------------|---------------------------------------|-----------|---------------------------------------------------------------------------------------|
| Condenser cleanliness factor (design)                   | 0.85 -           | HEI                                   | confirmed | ER-07.07 Condenser design standard                                                    |
| Condenser cooling water temperature rise                | 7 K              | maximum, base load                    | confirmed | ER-07.04 Cooling water temperature rise                                               |
| Condenser pressure at SRC (maximum)                     | 36 mbar(a)       | base load, seawater 16.0 degC         | confirmed | ER-07.08 Condenser pressure                                                           |
| Condenser tube velocity                                 | 1.8 to 2.2 m/s   | titanium                              | confirmed | ER-07.07 Condenser design standard                                                    |
| CW intake velocity at the screens                       | 0.15 m/s         | maximum, with fish return system      | confirmed | SRC-OWNR-0007 Positive EIA decision - binding conditions                              |
| Jellyfish load design                                   | 20 kg/m3         | blooms June-September                 | confirmed | SRC-OWNR-0002 Seawater quality and marine conditions, ER-08.07 Jellyfish and mucilage |
| Pycnocline depth                                        | 20 to 25 m       |                                       | confirmed | SRC-OWNR-0002 Seawater quality and marine conditions                                  |
| Residual chlorine at outfall                            | 0.1 mg/l         | maximum                               | confirmed | ER-08.04 Biofouling control                                                           |
| Seawater boron                                          | 2.4 to 3.4 mg/l  |                                       | confirmed | SRC-OWNR-0002 Seawater quality and marine conditions                                  |
| Seawater chloride - maximum (upper layer)               | 14,500 mg/l      |                                       | confirmed | SRC-OWNR-0002 Seawater quality and marine conditions                                  |
| Seawater salinity range (upper layer)                   | 21.8 to 26.1 PSU | lower layer 38.2 PSU below pycnocline | confirmed | SRC-OWNR-0002 Seawater quality and marine conditions                                  |
| Seawater temperature at intake - annual mean (measured) | 15.4 degC        | 12 m depth, station M2                | confirmed | SRC-OWNR-0002 Seawater quality and marine conditions                                  |
| Seawater temperature at intake - maximum                | 27 degC          | August                                | confirmed | SRC-OWNR-0002 Seawater quality and marine conditions                                  |
| Seawater temperature at intake - minimum                | 6.9 degC         | February                              | confirmed | SRC-OWNR-0002 Seawater quality and marine conditions                                  |
| Seawater temperature range at intake                    | 6 to 27 degC     |                                       | confirmed | ER-02.04 Seawater conditions                                                          |
| Seawater TSS maximum (storm)                            | 140 mg/l         |                                       | confirmed | SRC-OWNR-0002 Seawater quality and marine conditions                                  |
| SRC seawater temperature                                | 16 degC          | Site Reference Conditions             | confirmed | ER-02.03 Site Reference Conditions (SRC)                                              |
| SWRO brine salinity excess at 50 m from the diffuser    | 5 %              | above ambient salinity, maximum       | confirmed | SRC-OWNR-0007 Positive EIA decision - binding conditions                              |
| Thermal plume excess at 100 m mixing zone               | 3 K              | maximum                               | confirmed | ER-08.05 Thermal plume                                                                |
| Wave height Hs 50-year                                  | 4.4 m            | Tp 8.5 s, SW                          | confirmed | SRC-OWNR-0002 Seawater quality and marine conditions                                  |

## 9. Emissions and noise

Stack emission limits (15 % O2, dry) and stack height.

| Parameter | Value     | Condition                      | Status    | Basis                  |
|-----------|-----------|--------------------------------|-----------|------------------------|
| CO        | 30 mg/Nm3 | dry, 15 % O2, 50-100 % GT load | confirmed | ER-14.02 Air emissions |

| Parameter    | Value     | Condition                                | Status    | Basis                                                                            |
|--------------|-----------|------------------------------------------|-----------|----------------------------------------------------------------------------------|
| CO on LDO    | 50 mg/Nm3 | dry, 15 % O2, 70-100 % GT load           | confirmed | SRC-OWNR-0007 Positive EIA decision - binding conditions, ER-14.02 Air emissions |
| Dust on LDO  | 5 mg/Nm3  | dry, 15 % O2                             | confirmed | SRC-OWNR-0007 Positive EIA decision - binding conditions, ER-14.02 Air emissions |
| NH3 slip     | 5 mg/Nm3  | dry, 15 % O2                             | confirmed | ER-14.02 Air emissions                                                           |
| NOx (as NO2) | 30 mg/Nm3 | dry, 15 % O2, 50-100 % GT load           | confirmed | ER-14.02 Air emissions                                                           |
| NOx on LDO   | 50 mg/Nm3 | dry, 15 % O2                             | confirmed | SRC-OWNR-0007 Positive EIA decision - binding conditions, ER-14.02 Air emissions |
| SO2 on LDO   | 10 mg/Nm3 | dry, 15 % O2                             | confirmed | SRC-OWNR-0007 Positive EIA decision - binding conditions, ER-14.02 Air emissions |
| Stack height | 65 m      | EIA decision (approved dispersion study) | confirmed | ER-06.06 Stack, SRC-OWNR-0007 Positive EIA decision - binding conditions         |

Noise limits.

| Parameter                         | Value    | Condition              | Status    | Basis                                                                                  |
|-----------------------------------|----------|------------------------|-----------|----------------------------------------------------------------------------------------|
| Near-field noise                  | 85 dB(A) | 1 m from equipment     | confirmed | ER-14.03 Near-field noise                                                              |
| Noise at nearest receptor (day)   | 55 dB(A) | L <sub>Aeq</sub> day   | confirmed | SRC-OWNR-0007 Positive EIA decision - binding conditions, ER-14.04 Environmental noise |
| Noise at nearest receptor (night) | 45 dB(A) | L <sub>Aeq</sub> night | confirmed | ER-14.04 Environmental noise                                                           |

## 10. Guarantees

Owner guarantees (PG, contract ALP-EPC-001) and the power island guarantees of Imaginary Electric (IG, ALP-CA-001) that cover them back-to-back.

| Id    | Guarantee                                                                               | Value        | Min / max | Conditions                                                                                                                                                                                                             |
|-------|-----------------------------------------------------------------------------------------|--------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IG-01 | Power island gross output at the generator terminals                                    | 615 MW       | min       | Site Reference Conditions SRC-NG-100 (ER-02.03), new and clean, per IEC-ALP-PER-001 section 7                                                                                                                          |
| IG-02 | Power island gross heat rate (LHV)                                                      | 5,735 kJ/kWh | max       | Site Reference Conditions SRC-NG-100 (ER-02.03), new and clean, per IEC-ALP-PER-001 section 7                                                                                                                          |
| IG-03 | GT output at the generator terminals                                                    | 418 MW       | min       | Site Reference Conditions SRC-NG-100 (ER-02.03), new and clean, per IEC-ALP-PER-001 section 7                                                                                                                          |
| IG-04 | Condenser pressure at 16.0 degC seawater, 36,050-38,000 m3/h (corrected to 36,050 m3/h) | 35.5 mbar(a) | max       | SRC-NG-100, new and clean, ball cleaning in operation and debris filters clean, per IEC-ALP-PER-001 7                                                                                                                  |
| IG-05 | IEC auxiliary loads (excl. GSU losses)                                                  | 2,700 kW     | max       | Site Reference Conditions SRC-NG-100 (ER-02.03), new and clean, per IEC-ALP-PER-001 section 7                                                                                                                          |
| IG-06 | NOx at HRSG outlet (15 % O2, dry), 40-100 % GT load, natural gas                        | 30 mg/Nm3    | max       | Site Reference Conditions SRC-NG-100 (ER-02.03), new and clean, per IEC-ALP-PER-001 section 7                                                                                                                          |
| IG-07 | Power island gross heat rate (LHV) at the GT load for 60 % plant net load               | 6,060 kJ/kWh | max       | SRC, sliding pressure, new and clean, per IEC-ALP-PER-001 Rev A section 7                                                                                                                                              |
| IG-08 | GT minimum emissions-compliant load (natural gas)                                       | 30 % GT load | max       | Natural gas per ER-03.02, NOx and CO within the stack limits with SCR and oxidation catalyst                                                                                                                           |
| IG-09 | Power island gross output on LDO at the generator terminals                             | 572 MW       | min       | SRC, LDO with water/fuel 0.60, new and clean, per IEC-ALP-PER-003 Rev A case SRC-LDO-100                                                                                                                               |
| IG-10 | Power island gross heat rate (LHV) on LDO                                               | 6,020 kJ/kWh | max       | SRC, LDO with water/fuel 0.60, new and clean, per IEC-ALP-PER-003 Rev A case SRC-LDO-100                                                                                                                               |
| PG-01 | Net electrical output at Site Reference Conditions                                      | 600 MW       | min       | Base load, new and clean (corrected for recoverable degradation per agreed curves), Site Reference Conditions (ER-02.03), power factor 0.85 lagging at the 380 kV terminal point, all auxiliaries in normal operation. |
| PG-02 | Net heat rate (LHV) at Site Reference Conditions                                        | 5,860 kJ/kWh | max       | As PG-01, fuel within the Fuel Specification (ER-03), heat input on LHV basis.                                                                                                                                         |
| PG-03 | Net heat rate (LHV) at 60 % plant load                                                  | 6,250 kJ/kWh | max       | Plant at 60 % of net output, Site Reference Conditions, sliding pressure.                                                                                                                                              |
| PG-04 | NOx emission (as NO2)                                                                   | 30 mg/Nm3    | max       | Dry flue gas, 15 % O2, 50 % to 100 % GT load, all ambient conditions within the design range.                                                                                                                          |
| PG-05 | CO emission                                                                             | 30 mg/Nm3    | max       | Dry flue gas, 15 % O2, 50 % to 100 % GT load.                                                                                                                                                                          |
| PG-06 | Ammonia slip from the SCR                                                               | 5 mg/Nm3     | max       | Dry flue gas, 15 % O2, 50 % to 100 % GT load, catalyst new.                                                                                                                                                            |
| PG-07 | Near-field noise                                                                        | 85 dB(A)     | max       | 1 m from equipment surfaces / enclosures, 1.5 m height, all operating modes except emergency.                                                                                                                          |



| Id    | Guarantee                                                 | Value                      | Min / max | Conditions                                                                                                                                                                    |
|-------|-----------------------------------------------------------|----------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PG-08 | Noise at nearest sensitive receptor (night)               | 45 dB(A)                   | max       | LAeq at the nearest residential receptor defined in the EIA, plant at base load, night period.                                                                                |
| PG-09 | Hot start time to base load                               | 45 min                     | max       | Shutdown shorter than 8 h, GT ignition to plant base load, emissions compliant from minimum environmental load.                                                               |
| PG-10 | Load ramp rate in combined cycle                          | 35 MW/min                  | min       | Between minimum environmental load and base load, emissions compliant.                                                                                                        |
| PG-11 | Minimum environmental load                                | 40 % of plant net output   | max       | Lowest load with PG-04/PG-05 compliance at Site Reference Conditions.                                                                                                         |
| PG-12 | Reliability run availability                              | 98 %                       | min       | 30 consecutive days of operation following Owner dispatch, maximum two forced outages, none longer than 48 h.                                                                 |
| PG-13 | Condenser cooling water temperature rise                  | 7 K                        | max       | Base load, all seawater temperatures within the design range.                                                                                                                 |
| PG-14 | Thermal plume at edge of mixing zone                      | 3 K                        | max       | Temperature excess over ambient seawater at 100 m from the outfall diffuser, base load, summer conditions.                                                                    |
| PG-15 | Net electrical output on LDO at Site Reference Conditions | 560 MW                     | min       | Base load on the reference LDO (LHV 42.9 MJ/kg, FAME 0 %, water 100 mg/kg), water injection in service, SRC; corrections from the actual test fuel by agreed curves (TQ-023). |
| PG-16 | Net heat rate (LHV) on LDO at Site Reference Conditions   | 6,150 kJ/kWh               | max       | Base load on the reference LDO (LHV 42.9 MJ/kg, FAME 0 %, water 100 mg/kg), water injection in service, SRC; corrections from the actual test fuel by agreed curves (TQ-023). |
| PG-17 | NOx emission on LDO (as NO2)                              | 50 mg/Nm3                  | max       | Dry, 15 % O2, 50 % to 100 % GT load on LDO.                                                                                                                                   |
| PG-18 | Automatic fuel transfer gas to LDO without trip           | 100 % successful transfers | min       | Three transfers at 50 % load and three at base load, triggered by simulated loss of gas pressure.                                                                             |

## 11. Codes and standards

The applicable codes, standards and Turkish regulations, their editions and their order of precedence are listed in the Codes, Standards and Regulations Register (Appendix A.19, ALP-EPC-00000-GE-LST-0001); design to Eurocodes with Turkish national annexes and TBDY 2018 for structures, IEC for electrical and I&C, ASME / EN for pressure parts, NFPA 850 for fire protection.

## 12. Open items and assumptions

| Parameter                           | Value    | Status      | Basis                                                                                                           |
|-------------------------------------|----------|-------------|-----------------------------------------------------------------------------------------------------------------|
| Fire water design flow and pressure | 341 m3/h | preliminary | REF-CONS-0069 NFPA 850, REF-CONS-0070 NFPA 20, DEC-EPCE-0014 Preliminary equipment sizing basis and conventions |
| Fire water storage                  | 700 m3   | preliminary | REF-CONS-0069 NFPA 850, DEC-EPCE-0014 Preliminary equipment sizing basis and conventions                        |

## 13. Basis documents

| Reference               | Title                                                             |
|-------------------------|-------------------------------------------------------------------|
| decision:DEC-EPCE-0014  | DEC-EPCE-0014 Preliminary equipment sizing basis and conventions  |
| decision:DEC-OWNR-0002  | DEC-OWNR-0002 Acceptance tolerance of the Owner-levelled platform |
| reference:REF-CONS-0069 | REF-CONS-0069 NFPA 850                                            |
| reference:REF-CONS-0070 | REF-CONS-0070 NFPA 20                                             |
| reference:REF-CONS-0096 | REF-CONS-0096 TS EN 590                                           |
| requirement:ER-01.03    | ER-01.03 Design life and operating regime                         |
| requirement:ER-02.01    | ER-02.01 Site location and area                                   |
| requirement:ER-02.02    | ER-02.02 Ambient design range                                     |
| requirement:ER-02.03    | ER-02.03 Site Reference Conditions (SRC)                          |
| requirement:ER-02.04    | ER-02.04 Seawater conditions                                      |
| requirement:ER-02.05    | ER-02.05 Wind, snow and rain                                      |
| requirement:ER-02.06    | ER-02.06 Seismic design                                           |
| requirement:ER-02.08    | ER-02.08 Corrosive coastal atmosphere                             |
| requirement:ER-02.09    | ER-02.09 Flood protection                                         |
| requirement:ER-02.10    | ER-02.10 Airborne salt and dust                                   |
| requirement:ER-02.11    | ER-02.11 Design ambient points                                    |
| requirement:ER-02.12    | ER-02.12 Lightning                                                |
| requirement:ER-02.14    | ER-02.14 Access and transport                                     |
| requirement:ER-03.02    | ER-03.02 Natural gas specification                                |
| requirement:ER-03.03    | ER-03.03 Gas supply conditions at terminal point                  |
| requirement:ER-03.06    | ER-03.06 Natural gas contaminant limits                           |
| requirement:ER-03.07    | ER-03.07 Light diesel oil specification and burner limits         |
| requirement:ER-03.08    | ER-03.08 LDO unloading and storage                                |
| requirement:ER-04.01    | ER-04.01 Net electrical output                                    |
| requirement:ER-04.02    | ER-04.02 Net heat rate                                            |
| requirement:ER-04.03    | ER-04.03 Part-load efficiency                                     |

| Reference            | Title                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------|
| requirement:ER-04.04 | ER-04.04 Start-up times                                                                       |
| requirement:ER-04.05 | ER-04.05 Load ramp rate                                                                       |
| requirement:ER-04.06 | ER-04.06 Minimum environmental load                                                           |
| requirement:ER-04.07 | ER-04.07 Frequency and voltage support                                                        |
| requirement:ER-04.08 | ER-04.08 Reactive capability                                                                  |
| requirement:ER-04.10 | ER-04.10 Auxiliary power                                                                      |
| requirement:ER-06.06 | ER-06.06 Stack                                                                                |
| requirement:ER-07.04 | ER-07.04 Cooling water temperature rise                                                       |
| requirement:ER-07.07 | ER-07.07 Condenser design standard                                                            |
| requirement:ER-07.08 | ER-07.08 Condenser pressure                                                                   |
| requirement:ER-08.04 | ER-08.04 Biofouling control                                                                   |
| requirement:ER-08.05 | ER-08.05 Thermal plume                                                                        |
| requirement:ER-08.07 | ER-08.07 Jellyfish and mucilage                                                               |
| requirement:ER-11.01 | ER-11.01 Grid connection                                                                      |
| requirement:ER-14.02 | ER-14.02 Air emissions                                                                        |
| requirement:ER-14.03 | ER-14.03 Near-field noise                                                                     |
| requirement:ER-14.04 | ER-14.04 Environmental noise                                                                  |
| source:SRC-EPCE-0011 | SRC-EPCE-0011 Topographic survey point listing - plot, temporary area TP-A1 and coastal strip |
| source:SRC-OWNR-0001 | SRC-OWNR-0001 Meteorological and ambient air data                                             |
| source:SRC-OWNR-0002 | SRC-OWNR-0002 Seawater quality and marine conditions                                          |
| source:SRC-OWNR-0003 | SRC-OWNR-0003 Natural gas quality and supply conditions                                       |
| source:SRC-OWNR-0004 | SRC-OWNR-0004 Secondary fuel: light diesel oil (LDO)                                          |
| source:SRC-OWNR-0005 | SRC-OWNR-0005 Site, existing infrastructure and tie-in points                                 |
| source:SRC-OWNR-0006 | SRC-OWNR-0006 Grid connection data                                                            |
| source:SRC-OWNR-0007 | SRC-OWNR-0007 Positive EIA decision - binding conditions                                      |
| source:SRC-OWNR-0008 | SRC-OWNR-0008 Geotechnical baseline summary                                                   |